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GUIDE PIN FOR BLANK COVER DETECTION

Abstract

The present disclosure describes a supervisor card that uses guide pins on the backplane connectors in a chassis to determine whether covers are inserted in slots in the chassis. A system includes a backplane connector, a guide pin, a cover, and a processor. The backplane connector is positioned in a slot on a chassis. The guide pin is coupled to the backplane connector. The cover includes a metal clip that engages the guide pin when the cover is inserted into the slot. The processor is electrically coupled to the guide pin and applies a voltage to the guide pin. The processor also determines that the cover is inserted into the slot based on a change in voltage or current in the guide pin.

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Background/Summary

TECHNICAL FIELD

[0001] Embodiments presented in this disclosure generally relate to hardware racks. More specifically, embodiments disclosed herein a guide pin used to detect blank covers in hardware racks.

BACKGROUND

[0002] Hardware racks may be used to hold hardware devices (e.g., in a datacenter or server room). For example, a hardware rack may hold line cards, supervisor cards, etc. in slots in a chassis of the rack. When a slot in the chassis is not being used, a cover (which may be referred to as a blank cover) may be inserted into the slot to improve thermal performance and to protect other hardware devices in the chassis. In existing systems, the supervisor cards may detect when the cover is inserted in the slot using a Hall detector positioned on the chassis and near the cover. The Hall detector, however, occupies additional space in the rack and increases the complexity of the design.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate typical embodiments and are therefore not to be considered limiting; other equally effective embodiments are contemplated.

[0004] FIG. 1A illustrates an example system.

[0005] FIG. 1B illustrates components of an example supervisor card in the system of FIG. 1A.

[0006] FIG. 2A through 2F illustrate example covers and portions of the covers in the system of FIG. 1A.

[0007] FIG. 3 illustrates an example portion of the system of FIG. 1A.

[0008] FIGS. 4A through 4D illustrate example portions of the system of FIG. 1A.

[0009] FIGS. 5A and 5B illustrate example portions of the system of FIG. 1A.

[0010] FIG. 6 illustrates an example portion of the system of FIG. 1A.

[0011] FIG. 7 is a flowchart of an example method performed by the system of FIG. 1A.

[0012] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially used in other embodiments without specific recitation.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

[0013] The present disclosure describes a supervisor card that uses guide pins on the backplane connectors in a chassis to determine whether covers are inserted in slots in the chassis. According to an embodiment, a system includes a backplane connector, a guide pin, a cover, and a processor. The backplane connector is positioned in a slot on a chassis. The guide pin is coupled to the backplane connector. The cover includes a metal clip that engages the guide pin when the cover is inserted into the slot. The processor is electrically coupled to the guide pin and applies a voltage to the guide pin. The processor also determines that the cover is inserted into the slot based on a change in voltage or current in the guide pin.

[0014] According to another embodiment, a method includes applying a voltage to a guide pin coupled to a backplane connector positioned in a slot on a chassis. The method also includes determining that a cover comprising a metal clip is inserted into the slot based on a change in voltage or current in the guide pin. The metal clip engages the guide pin when the cover is inserted

into the slot.

[0015] According to another embodiment, a system includes a first guide pin, a second guide pin, a cover, a line card, and a processor. The first guide pin is positioned in a first slot on a chassis. The second guide pin is positioned in a second slot on the chassis. The cover includes a metal clip. The cover is inserted into the first slot such that the metal clip engages the first guide pin. The line card includes a plastic connector. The line card is inserted into the second slot such that the plastic connector engages the second guide pin. The processor is electrically coupled to the first guide pin and the second guide pin. The processor applies a voltage to the first guide pin and the second guide pin and determines that the cover is inserted into the first slot rather than the second slot based on a change in voltage or current in the guide pin.

Example Embodiments

[0016] The present disclosure describes a supervisor card that uses guide pins in a chassis to determine whether covers are inserted in slots in the chassis. For example, the supervisor card may apply an electric voltage to the guide pins on the backplane connectors in the chassis. A cover may include a metal clip that engages a guide pin when the cover is inserted in a slot. The metal clip causes a change in voltage or current in the guide pin. For example, the metal clip may pull the guide pin to electrical ground and cause an electric current to flow through the guide pin. When the supervisor card detects the change in voltage or current, the supervisor card may determine that a cover is inserted in the slot. If, on the other hand, a line card were inserted in the slot, the line card may not include the metal clip that engages the guide pin. As a result, when the line card is inserted in the slot, the line card may not pull the guide pin to electrical ground. The supervisor card may determine that the cover is not inserted in the slot when the supervisor card detects that the guide pin has not been pulled to electrical ground.

[0017] In certain embodiments, the supervisor card and chassis provide several technical advantages. For example, by using the guide pins to determine whether covers are inserted into slots in the chassis, the amount of real estate in the chassis occupied by hardware is reduced relative to existing systems that use Hall detectors. As another example, by using the guide pins, the complexity and cost of designs is reduced relative to existing systems that use Hall detectors.

[0018] FIG. 1A illustrates an example system **100**. The system **100** may be a hardware rack (e.g., in a data center or server room). As seen in FIG. 1A, the system **100** includes a chassis **102**. The chassis **102** may be a structure (e.g., metal structure, plastic structure, wood structure, etc.) that holds or houses multiple hardware devices. The chassis **102** may hold or house hardware devices in any configuration. In the example of FIG. 1A, the chassis **102** holds or houses multiple hardware devices arranged vertically in the chassis **102**.

[0019] The chassis **102** may store or house any type of hardware device. Typically, the hardware devices may be arranged on one or more line cards **104** that are inserted and secured to the chassis **102**. Each of the line cards **104** may include a different hardware device that performs a different function. The chassis **102** may also store or hold one or more supervisor cards **106**. The supervisor cards **106** may include hardware devices that monitor or control the line cards **104**. In the example of FIG. 1A, the chassis **102** holds or houses eight line cards **104** and two supervisor cards **106**. The line cards **104** may be electrically and communicatively connected to the supervisor cards **106** by backplane connectors positioned in the chassis **102**. When the line cards **104** and the supervisor cards **106** are positioned into slots in the chassis **102**, the line cards **104** and the supervisor cards **106** may physically connect to the backplane connectors in the chassis **102**. These backplane connectors may then connect the line cards **104** and the supervisor cards **106**.

[0020] When a slot is unused (e.g., when no line card **104** or supervisor card **106** is positioned in the slot), a cover may instead be positioned in the slot to improve thermal performance and to protect other components in the chassis **102** (e.g., other line cards **104** or supervisor cards **106**). The backplane connectors may include a metal (e.g., iron) guide pin that extends from the backplane connector. The guide pin may assist in aligning the line cards **104** and the supervisor cards **106** with

their respective backplane connectors. In certain embodiments, the supervisor cards **106** apply an electric voltage to the guide pins in the backplane connectors. When a cover is positioned in a slot, the cover may cause a change in voltage or current in the in the guide pin for that slot (e.g., pull the guide pin to electrical ground). When a line card **104** or supervisor card **106** is positioned in the slot, the guide pin may not be pulled to electrical ground. The supervisor cards **106** may monitor the voltages or currents across the guide pins to determine whether covers are inserted into the slots in the chassis **102**.

[0021] FIG. **1B** illustrates an example line card **104** and/or supervisor card **106** in the system **100** of FIG. **1A**. As discussed previously, the line card **104** and the supervisor card **106** may be positioned in slots in the chassis **102**. As seen in FIG. **1B**, the line card **104** and/or the supervisor card **106** may include a processor **108** and a memory **110** that perform any of the functions of the line card **104** and/or the supervisor card **106** described herein. Each line card **104** and each supervisor card **106** in the chassis **102** may include a processor **108** and a memory **110**.

[0022] The processor **108** is any electronic circuitry, including, but not limited to one or a combination of microprocessors, microcontrollers, application specific integrated circuits (ASIC), application specific instruction set processor (ASIP), and/or state machines, that communicatively couples to the memory **110** and controls the operation of the line card **104** and/or the supervisor card **106**. The processor **108** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **108** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The processor **108** may include other hardware that operates software to control and process information. The processor **108** executes software stored on the memory **110** to perform any of the functions described herein. The processor **108** controls the operation and administration of the line card **104** and/or the supervisor card **106** by processing information (e.g., information received from the line cards **104**, supervisor cards **106**, and memory **110**). The processor **108** is not limited to a single processing device and may encompass multiple processing devices contained in the same device or computer or distributed across multiple devices or computers. The processor **108** is considered to perform a set of functions or actions if the multiple processing devices collectively perform the set of functions or actions, even if different processing devices perform different functions or actions in the set.

[0023] The memory **110** may store, either permanently or temporarily, data, operational software, or other information for the processor **108**. The memory **110** may include any one or a combination of volatile or non-volatile local or remote devices suitable for storing information. For example, the memory **110** may include random access memory (RAM), read only memory (ROM), magnetic storage devices, optical storage devices, or any other suitable information storage device or a combination of these devices. The software represents any suitable set of instructions, logic, or code embodied in a computer-readable storage medium. For example, the software may be embodied in the memory **110**, a disk, a CD, or a flash drive. In particular embodiments, the software may include an application executable by the processor **108** to perform one or more of the functions described herein. The memory **110** is not limited to a single memory and may encompass multiple memories contained in the same device or computer or distributed across multiple devices or computers. The memory **110** is considered to store a set of data, operational software, or information if the multiple memories collectively store the set of data, operational software, or information, even if different memories store different portions of the data, operational software, or information in the set.

[0024] FIGS. **2A** through **2F** illustrate example covers **202** and **220** and portions of the cover **202** and **220** in the system **100** of FIG. **1A**. Generally, when a slot in the chassis **102** is not holding or housing a hardware device (e.g., a line card **104** or a supervisor card **106**), a cover **202** or **220** is

instead inserted into the slot. The cover **202** or **220** may be a metallic tray that occupies the empty slot. In certain embodiments, the cover **202** or **220** improves thermal performance and protects other hardware devices in the chassis **102**.

[0025] To understand how to control and manage the devices in the chassis **102**, the supervisor cards **106** may need to know which slots are occupied by covers **202** or **220**. In existing systems, a Hall detector may be positioned on the chassis and near the cover to detect when the cover is inserted in the slot. The Hall detector, however, occupies additional space in the chassis, and increasing the complexity of the design. The present disclosure describes a cover **202** or **220** that includes a metal clip that engages the guide pin of a backplane connector when the cover **202** or **220** is inserted into a slot in the chassis **102**. The metal clip may cause a change in voltage or current in the guide pin (e.g., pull the guide pin to electrical ground). Because the supervisor card **106** is applying an electrical voltage to the guide pin, the metal clip may complete a circuit and cause an electric signal or current to flow through the guide pin. When the supervisor card **106** detects the change in voltage or current in the guide pin, the supervisor card **106** may determine that the cover **202** or **220** is inserted into the slot in the chassis **102**. For example, the supervisor card **106** may monitor the voltage across the guide pin to determine when the guide pin has been pulled to electrical ground. Alternatively or additionally, the supervisor card **106** may monitor the electrical signals or currents flowing through the guide pin. When the supervisor card **106** detects an electrical signal or current flowing through the guide pin, the supervisor card **106** may determine that the cover **202** or **220** is inserted into the slot in the chassis **102**.

[0026] When a line card **104** is inserted into the slot, a plastic mating component may engage the guide pin. The plastic mating component may not complete the circuit, and thus, the supervisor card **106** may not detect a change in voltage or current in the guide pin. In response, the supervisor card **106** may determine that a cover **202** or **220** is not inserted into the slot in the chassis **102**. In some embodiments, when the line card **104** is inserted into the slot in the chassis **102**, the line card **104** causes an electric signal to flow through another pin (e.g., an electrical pin on a backplane of the chassis **102**). When the supervisor card **106** detects this electric signal (and/or no electric signal flowing through the guide pin), the supervisor card **106** may determine that the line card **104** is inserted into the slot.

[0027] FIG. 2A illustrates an example cover **202**. As seen in FIG. 2A, the cover **202** may be a metallic tray that fits within a slot in the chassis **102**. The cover **202** includes a front side **204** and a backside **206**. The backside **206** of the cover **202** may first be inserted through a front side of a slot in the chassis **102**. The backside **206** may then be pushed towards a backside of the slot in the chassis **102**. When the cover **202** is positioned in the slot, the backside **206** of the cover **202** is positioned near the backside of the chassis **102** and the front side **204** is positioned near the front side of the chassis **102**. The backplane connector for the slot may also be positioned on the backside of the chassis. As a result, the backside **206** of the cover **202** is positioned near the backplane connector for the slot when the cover **202** is positioned in the slot.

[0028] Additionally, as seen in FIG. 2A, the cover **202** defines multiple cavities **208** in the backside **206**. The cavities **208** may provide improved airflow through the cover **202**. The backside **206** may define any number of cavities **208**. In the example of FIG. 2A, the backside **206** defines a single row of cavities **208**. Due to the height of the cover **202**, the cover **202** may be suitable for slots that are sized to hold or house line cards **104**.

[0029] FIG. 2B illustrates another perspective of the cover **202**. As seen in FIG. 2B, the cover **202** includes the front side **204** and the backside **206** opposite the front side **204**. The backside **206** also defines the cavities **208**. Additionally, a metal clip **210** is positioned on the backside **206**. As discussed previously, when the cover **202** is positioned in the slot of the chassis **102**, the clip **210** may engage a guide pin of the backplane connector for the slot. The clip **210** may cause a change in voltage or current in the guide pin (e.g., pull the guide pin to electrical ground). Because a supervisor card **106** may apply an electrical voltage to the guide pin, the clip **210** may cause an

electrical signal or current to flow through the guide pin when the clip **210** engages the guide pin. The supervisor card **106** may detect the change in voltage or current in the guide pin to determine that the cover **202** (rather than a line card) is positioned in the slot.

[0030] FIG. 2C illustrates the clip **210**. As seen in FIG. 2C, the clip **210** may be secured to the backside of **206** of the cover **202**. The clip **210** may include arms **212** and **214**. The arms **212** and **214** may open and close around a guide pin of a backplane connector to engage the guide pin. Additionally, the arms **214** and **212** may pull the guide pin to electrical ground, which may cause an electric signal or current to flow through the guide pin.

[0031] FIG. 2D illustrates another perspective of the clip **210**. As seen in FIG. 2D, the clip **210** includes the arms **212** and **214**. The arms **212** and **214** may be connected or joined together at one end of the arms **212** and **214**. The other end of the arms **212** and **214** may open or close around a guide pin to engage the guide pin.

[0032] FIG. 2E illustrates a cover **220**. As seen in FIG. 2E, the cover **220** may be larger than the cover **202**. As a result, the cover **220** may be designed for a large slot in the chassis **102**. For example, the cover **202** may be designed for a slot that holds a line card **104**, while the cover **220** is designed for a slot that holds a supervisor card **106**.

[0033] The cover **220** includes the front side **204** and the backside **206** opposite the front side **204**. Additionally, the backside **206** defines multiple cavities **208**. As seen in FIG. 2E, the backside **206** defines an additional row of cavities **208** relative to the cover **202**. The cover **220** may be inserted into a slot in the chassis **102** by first inserting the backside **206** into the slot through the front side of the slot. The backside **206** may then be pushed towards the backside of the slot. When the cover **220** is positioned in the slot, the backside **206** may be positioned near the backside of the chassis **102**. Additionally, the front side **204** may be positioned near the front side of the chassis **102**.

[0034] FIG. 2F illustrates another perspective of the cover **220**. As seen in FIG. 2F, the cover **220** includes the front side **204** and the backside **206** opposite the front side **204**. The backside **206** defines multiple rows of cavities **208**. Additionally, the clip **210** is attached to the backside **206**. The clip **210** may engage a guide pin of a backplane connector for a slot when the cover **220** is positioned in the slot. The cover **220** may cause a change in voltage or current in the guide pin. For example, the cover **220** may pull the guide pin to electrical ground, which causes an electric signal or current to flow through the guide pin. A supervisor card **106** in the chassis **102** may determine that the cover **220** (rather than a supervisor card) is positioned in the slot when the supervisor card **106** detects that the guide pin has been pulled to electrical ground.

[0035] FIG. 3 illustrates an example portion of the system **100** of FIG. 1A. As seen in FIG. 3, the system **100** includes a printed circuit board **302**. Multiple backplane connectors **304** are attached to the printed circuit board **302**. The printed circuit board **302** may include electrical connections that electrically connect the backplane connectors **304** to each other. The printed circuit board **302** may be positioned along a backside of the chassis **102**. The printed circuit board **302** may be positioned such that each backplane connector **304** is positioned in a slot of the chassis **102**. When a hardware device (e.g., line card **104** or supervisor card **106**) or cover **202** or **220** is inserted into the slot, the hardware device or cover **202** or **220** may engage the backplane connector **304** for that slot.

[0036] As seen in FIG. 3, the printed circuit board **302** includes four backplane connectors **304A** arranged in an upper portion of the printed circuit board **302**. The printed circuit board **302** also includes two backplane connectors **304B** arranged in a middle portion of the printed circuit board **302** (e.g., between the upper and lower portions). Additionally, the printed circuit board **302** includes four backplane connectors **304A** arranged in a lower portion of the printed circuit board **302**. The four backplane connectors **304A** arranged in the upper portion of the printed circuit board **302** and the four backplane connectors **304A** arranged in the lower portion of the printed circuit board **302** may be positioned in slots that accommodate line cards **104**. When the line cards **104** are positioned in these slots, the line cards **104** may connect to these backplane connectors **304A**. The backplane connectors **304B** positioned in the middle portion of the printed circuit board **302** may

be positioned in slots that accommodate supervisor cards **106**. When supervisor cards **106** are positioned in the slots, the supervisor cards **106** connect to these backplane connectors **304B**. [0037] FIGS. **4A** through **4D** illustrate example portions of the system **100** of FIG. **1A**. As seen in FIG. **4A**, the backplane connector **304** is attached to the printed circuit board **302**. Additionally, the backplane connector **304** includes a guide pin **306** that extends out from the backplane connector **304**. The guide pin **306** may be a metallic (e.g., iron) guide pin **306**. As seen in FIG. **4B**, the backplane connector **304** and the guide pin **306** may be secured to the printed circuit board **302** by a screw **308**. The screw **308** may extend through the printed circuit board **302** and the backplane connector **304**. The screw **308** may secure the backplane connector **304** to the printed circuit board **302**. Additionally, the screw **308** may extend into the guide pin **306** to secure the guide pin **306** to the backplane connector **304** and the printed circuit board **302**. As seen in FIG. **4C**, the screw **308** may engage a backside of the backplane connector **304**. The screw **308** may extend through the backplane connector **304** and into the guide pin **306**. In this manner, the screw **308** secures the backplane connector **304** and the guide pin **306** to the printed circuit board **302**.

[0038] FIG. **4D** illustrates another perspective of the backplane connector **304**. As seen in FIG. **4D**, the backplane connector **304** may include the guide pin **306**. The guide pin **306** may extend out of the backplane connector **304**. When a line card **104**, supervisor card **106**, or cover **202** or **220** engages the backplane connector **304**, the line card **104**, supervisor card **106**, or cover **202** or **220** may also engage the guide pin **306**. The guide pin **306** may assist in properly aligning the line card **104**, supervisor card **106**, or cover **202** or **220** with the backplane connector **304**. The cover **202** or **220** includes the clip **210** that engages the guide pin **306** when the cover **202** or **220** is positioned in the slot. When a line card **104** or supervisor card **106** engages the backplane connector **304**, the guide pin **306** may assist in aligning the line card **104** or supervisor card **106** with the backplane connector **304** so that the proper electrical connections are formed between the backplane connector **304** and the line card **104** or supervisor card **106**.

[0039] FIGS. **5A** and **5B** illustrate example portions of the system **100** of FIG. **1A**. As seen in FIG. **5A**, the chassis **102** includes one or more slots **502**. In the example of FIG. **5A**, the slots **502A**, **502B**, and **502C** are sized for line cards **104**. The slots **502D** and **502E** are sized for supervisor cards **106**. Line cards **104** are positioned in the slots **502A** and **502C**. Additionally, a supervisor card **106** is positioned in the slot **502E**. The printed circuit board **302** is positioned near a backside of the chassis **102**. When the line cards **104** and the supervisor card **106** are positioned in the slots **502A**, **502C**, and **502E**, the line cards **104** and the supervisor card **106** engage with the backplane connectors **304** positioned on the printed circuit board **302**. The supervisor card **106** may communicate with the line cards **104** through these backplane connectors **304**.

[0040] Additionally, a cover **202** is positioned in the slot **502B**. The cover **202** includes the clip **210** that engages with the guide pin **306** extending from the backplane connector for the slot **502B**. The supervisor card **106** may apply an electric voltage to the guide pin **306**. When the clip **210** on the cover **202** engages with the guide pin **306**, the clip **210** causes a change in voltage or current in the guide pin **306** (e.g., pulls the guide pin **306** to electrical ground). As a result, an electric signal or current flows through the guide pin **306** when the cover **202** is positioned in the slot **502B**. When the supervisor card **106** detects the change in voltage or current in the guide pin **306**, the supervisor card **106** may determine that the cover **202** is positioned in the slot **502B** and that no line card **104** is positioned in the slot **502B**.

[0041] FIG. **5B** shows a view of the backplane connector **304** and the cover **202** in the slot **502B**. As seen in FIG. **5B**, the printed circuit board **302** and the backplane connector **304** are positioned near a backside of the slot **502B**. When the cover **202** is positioned in the slot **502B**, the clip **210** on the backside **206** of the cover **202** engages the guide pin **306** extending from the backplane connector **304**. The clip **210** may cause a change in voltage or current in the guide pin **306**, which causes an electrical signal or current to flow through the guide pin **306** when the supervisor card **106** applies an electrical voltage to the guide pin **306** (e.g., through the backplane connectors **304**

for the slots **502B** and **502E**). When the supervisor card **106** detects the change in voltage or current in the guide pin **306** (e.g., by monitoring the voltage across the guide pin **306**), the supervisor card **106** may determine that the cover **202** is positioned in the slot **502B**. In some embodiments, a resistor may be coupled in series with the guide pin **306**. The resistor may prevent a short circuit when the cover **202** is positioned in the slot **502B**.

[0042] FIG. **6** illustrates an example portion of the system **100** of FIG. **1A**. Generally, FIG. **6** shows the backside of the chassis **102**. As seen in FIG. **6**, the line card **104** includes a connector **602** near a backside of the line card **104**. The connector **602** may engage a backplane connector in a slot of the chassis **102** when the line card **104** is inserted into the slot. The connector **602** may then communicate signals between the line card **104** and a supervisor card **106** through the backplane connector. Additionally, as seen in FIG. **6**, the connector **602** defines a cavity **604**. The cavity **604** may receive or engage the guide pin extending from the backplane connector. As a result, the cavity **604** and the guide pin align the connector **602** and the line card **104** with the backplane connector. In certain embodiments, the portion of the connector **602** that forms the cavity **604** may be formed using a plastic material or another electrically non-conducting material. As a result, then the cavity **604** receives or engages the guide pin, the connector **602** may not cause a change in voltage or current in the guide pin (e.g., does not pull the guide pin to ground). When a supervisor card applies electric voltage to the guide pin, no electric signal or current may flow through the guide pin due to the plastic or other non-conducting material engaging the guide pin. When the supervisor card determines that no electric signal or current is flowing through the guide pin, the supervisor card may determine that a cover **202** or **220** is not positioned in the slot **502**.

[0043] FIG. **7** is a flowchart of an example method **700** performed by the system **100** of FIG. **1A**. In particular embodiments, a supervisor card **106** in a chassis **102** may perform the method **700**. By performing the method **700**, the supervisor card **106** detects when a cover **202** or **220** is positioned in a slot **502** of the chassis **102**.

[0044] In block **702**, the supervisor card **106** applies a voltage to a guide pin **306** of a backplane connector **304** for a slot **502** in the chassis **102**. The supervisor card **106** may be connected to a backplane connector **304** that is connected to the backplane connector **304** for the slot **502** (e.g., through the printed circuit board **302**). The supervisor card **106** may apply the electric voltage through the backplane connectors **304** into the guide pin **306**.

[0045] In block **704**, the supervisor card **106** detects a change in voltage or current in the guide pin **306** (e.g., detect that the guide pin **306** is pulled to electrical ground). Generally, when a line card **104** is positioned in the slot **502**, the line card **104** may engage the guide pin **306** with a plastic or other non-conducting material. As a result, no change in voltage or current in the guide pin **306** occurs. On the other hand, when a cover **202** or **220** is positioned in the slot **502**, the metal clip **210** on the cover **202** or **220** engages the guide pin **306**, which causes a change in voltage or current in the guide pin **306** (e.g., pulls the guide pin **306** to electrical ground). As a result, an electric current flows through the guide pin **306** when the supervisor card **106** applies the electric voltage to the guide pin **306**.

[0046] In block **706**, the supervisor card **106** determines that the cover **202** and **220** is inserted in the slot **502** when the supervisor card **106** determines the change in voltage or current in the guide pin **306**. The supervisor card **106** may then log that the slot **502** is unused.

[0047] In summary, a supervisor card **106** uses guide pins **306** in a chassis **102** to determine whether covers **202** or **220** are inserted in slots **502** in the chassis **102**. For example, the supervisor card **106** may apply an electric voltage to the guide pins **306** on the backplane connectors **304** in the chassis **102**. A cover **202** or **220** may include a metal clip **210** that engages a guide pin **306** when the cover **202** or **220** is inserted in a slot **502**. The metal clip **210** may cause a change in voltage or current in the guide pin **306** (e.g., pulls the guide pin **306** to ground and causes an electric current to flow through the guide pin **306**). When the supervisor card **106** detects the change in voltage or current in the guide **306**, the supervisor card **106** may determine that a cover

202 or **220** is inserted in the slot **502**. If, on the other hand, a line card **104** were inserted in the slot **502**, the line card **104** may not include the metal clip **210** that engages the guide pin **306**. As a result, when the line card **104** is inserted in the slot **502**, the line card **104** may not cause a change in voltage or current in the guide pin **306**. The supervisor card **106** may determine that the cover **202** or **220** is not inserted in the slot **502** when the supervisor card **106** does not detect a change in voltage or current in the guide pin **306**.

[0048] In the current disclosure, reference is made to various embodiments. However, the scope of the present disclosure is not limited to specific described embodiments. Instead, any combination of the described features and elements, whether related to different embodiments or not, is contemplated to implement and practice contemplated embodiments. Additionally, when elements of the embodiments are described in the form of “at least one of A and B,” or “at least one of A or B,” it will be understood that embodiments including element A exclusively, including element B exclusively, and including element A and B are each contemplated. Furthermore, although some embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the scope of the present disclosure. Thus, the aspects, features, embodiments and advantages disclosed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the invention” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

[0049] As will be appreciated by one skilled in the art, the embodiments disclosed herein may be embodied as a system, method or computer program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.”

Furthermore, embodiments may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

[0050] Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0051] Computer program code for carrying out operations for embodiments of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0052] Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems), and computer program products according to embodiments presented in this disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts

specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0053] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0054] The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process such that the instructions which execute on the computer, other programmable data processing apparatus, or other device provide processes for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0055] The flowchart illustrations and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowchart illustrations or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0056] In view of the foregoing, the scope of the present disclosure is determined by the claims that follow.

Claims

1. A system comprising: a backplane connector positioned in a slot on a chassis; a guide pin coupled to the backplane connector; a cover comprising a metal clip, wherein the metal clip is arranged to engage the guide pin when the cover is inserted into the slot; and a processor electrically coupled to the guide pin, wherein the processor is configured to: apply a voltage to the guide pin; and determine that the cover is inserted into the slot based on a change in voltage or current in the guide pin.
2. The system of claim 1, wherein the metal clip pulls the guide pin to electrical ground when the metal clip engages the guide pin.
3. The system of claim 1, further comprising a line card comprising a plastic connector arranged to engage the guide pin when the line card is inserted into the slot.
4. The system of claim 1, further comprising a printed circuit board, wherein the backplane connector is attached to the printed circuit board.
5. The system of claim 4, further comprising a screw that attaches (i) the guide pin to the backplane connector and (ii) the backplane connector to the printed circuit board.
6. The system of claim 1, wherein the backplane connector is positioned on a backside of the chassis, and wherein the cover is arranged to insert into the slot through a front side of the chassis opposite the backside.
7. The system of claim 1, wherein the clip is coupled to a backside of the cover, and wherein the backside of the cover defines a plurality of cavities.
8. The system of claim 1, wherein determining that the cover is inserted into the slot is based on an

electric current flowing through the guide pin.

9. A method comprising: applying a voltage to a guide pin coupled to a backplane connector positioned in a slot on a chassis; and determining that a cover comprising a metal clip is inserted into the slot based on a change in voltage or current in the guide pin, wherein the metal clip engages the guide pin when the cover is inserted into the slot.

10. The method of claim 9, wherein the metal clip pulls the guide pin to electrical ground when the metal clip engages the guide pin.

11. The method of claim 9, wherein a line card comprising a plastic connector that engages the guide pin when the line card is inserted into the slot.

12. The method of claim 9, wherein the backplane connector is attached to a printed circuit board.

13. The method of claim 12, wherein a screw attaches (i) the guide pin to the backplane connector and (ii) the backplane connector to the printed circuit board.

14. The method of claim 9, wherein the backplane connector is positioned on a backside of the chassis, and wherein the cover is arranged to insert into the slot through a front side of the chassis opposite the backside.

15. The method of claim 9, wherein the clip is coupled to a backside of the cover, and wherein the backside of the cover defines a plurality of cavities.

16. A system comprising: a first guide pin positioned in a first slot on a chassis; a second guide pin positioned in a second slot on the chassis; a cover comprising a metal clip, wherein the cover is inserted into the first slot such that the metal clip engages the first guide pin; a line card comprising a plastic connector, wherein the line card is inserted into the second slot such that the plastic connector engages the second guide pin; and a processor electrically coupled to the first guide pin and the second guide pin, wherein the processor is configured to: apply a voltage to the first guide pin and the second guide pin; and determine that the cover is inserted into the first slot rather than the second slot based on a change in voltage or current in the first guide pin.

17. The system of claim 16, wherein the metal clip pulls the first guide pin to electrical ground when the metal clip engages the first guide pin.

18. The system of claim 16, further comprising backplane connector coupled to the first guide pin and the second guide pin.

19. The system of claim 18, further comprising a printed circuit board, wherein the backplane connector is attached to the printed circuit board.

20. The system of claim 19, further comprising a screw that attaches (i) the first guide pin to the backplane connector and (ii) the backplane connector to the printed circuit board.
